Blockchains

Blockchain

An append only ledger.

- An immutable record of all transactions and actions performed in the system so far.
- Each page of this ledger is called a block.
- These pages are "glued" together via a hash chain.
- Adding a new page on the ledger is done through a process called mining

Bitcoin in a Nutshell

A distributed currency exchange medium open for anyone to join, powered by a P2P network.
Utilize basic cryptographic primitives to control the money flow in the system (hash functions and digital signatures)
Building blocks:

- Players: miners and clients.
- Transaction: messages exchanged
- Blockchain: an append only log
- Mining: extending the blockchain
- Concensus: agreeing on the current state of the Blockchain

The Blockchain Structure

Header and Transaction:

1. Header: H(prev-block) (hash of the previous block's header, hash chain)
2. H(transactions) (root of Merkle Tree)
3. Difficulty/nonce/... (difficulty of mining)
4. Transactions
5. Coinbase: special transactions with no inputs and a single output to the miner

Mining

The miners extend the blockchain with new blocks (and mint new currency)
Done through proof-of-work (needed to prevent Sybil attacks)
Miners solve a hash puzzle, SHA-256(SHA-256(new block header)) ¡ Difficulty Target
For secure hash functions, the only way to find the hash with a given property is to try nonce values until a desired hash is found (solving a hash puzzle)
Verification is very easy, other miners check the validity of all transactions in a block, and then verify the solution of the hash puzzle (the latter is a single hash invocation)

Required Security Properties

Need one-way function so you can't reverse the difficulty target.